

# CLIMATE CHANGE AND SUBMERGED HERITAGE IN THE NORTH SEA:

## Assessment of Accelerating Threats to Organic Wreck Preservation

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## Executive Summary

The North Sea contains one of the highest concentrations of archaeologically significant underwater sites in the world, spanning more than two millennia of maritime activity. A substantial proportion of these sites preserve organic material, including ship timbers, rigging, textiles, and botanical cargo, under conditions that have maintained anaerobic sediment chemistry for centuries or longer. This working paper argues that climate-driven changes to North Sea temperature, storm frequency, and sediment dynamics are now measurably accelerating the degradation of this organic preservation environment and that without systematic documentation and strategic intervention, a large proportion of the surviving organic wreck record may be lost within one to two generations.

## 1. Background

The preservation of organic archaeological material in the North Sea and adjacent shelf waters depends primarily on stable, anaerobic sediment conditions. Where fine-grained sediment covers wreck deposits to a depth of 30 centimetres or more, oxygen penetration is limited, bacterial activity is reduced, and cellulose-based materials such as timber and rope can survive for hundreds of years with relatively low degradation rates. This preservation regime has been documented at numerous sites across the region, from the Wadden Sea and the German Bight to the deeper waters of the central North Sea basin.

## 2. Observed Changes to the Preservation Environment

Analysis of historical oceanographic datasets covering the period 1970 to 2024 reveals three categories of change relevant to organic site preservation:

2.1 Temperature increase: Mean annual sea surface temperature in the German Bight has increased by approximately 1.7 degrees Celsius over the 1970-2024 period, with the rate of increase accelerating in the most recent decade. Bottom water temperatures at 10-15 metre depth, the range relevant to most Wadden Sea heritage sites, show a comparable trend with less seasonal variability. Elevated temperature increases the metabolic rates of aerobic and anaerobic decomposition bacteria, shortening the effective preservation horizon even in stable sediment conditions.

2.2 Storm intensification: Instrumental records and modelling studies indicate a statistically significant increase in the frequency of severe storm events (defined as sustained winds above 40 knots at the German Bight weather stations) over the 1980-2024 period. Severe storms generate bottom currents sufficient to remobilise surface sediment at typical Wadden Sea wreck depths, exposing organic material to oxygenated water for days to weeks at a time. At Site NF-7, three

storm events between 2020 and 2024 caused measurable changes to the exposed timber distribution, confirmed by comparison of repeat multibeam surveys.

2.3 Acidification: Mean pH in the German Bight surface waters has declined by 0.12 units between 1990 and 2024, consistent with global ocean acidification trends. The effect of reduced pH on submerged organic material is less well studied than the effect on calciferous marine organisms, but preliminary data from the project sediment sampling programme suggest that the buffering capacity of the Wadden Sea sediments may be declining.

### 3. Site NF-7 as a Case Study

Site NF-7, the primary investigation site of the North Frisian Medieval Wreck Survey, provides a concrete illustration of the processes described above. The site was initially detected in 2020 by comparison of archival sidescan sonar data collected in 1998 with new multibeam survey results. The comparison revealed that a timber feature visible in the 1998 data as a buried deposit had become partially exposed by 2020, with visible hull timbers extending above the sediment surface in an area of approximately 15 square metres.

Repeat surveys in 2023 and 2025 show further exposure and some physical displacement of timber fragments, most likely due to the storm event of December 2024. The exposure trend is consistent with the broader pattern of sediment thinning observed in the outer Wattenmeer transition zone, which is itself linked to reduced sediment supply from eroding barrier island systems.

### 4. Implications for Archaeological Research Strategy

The evidence presented here has direct implications for the prioritisation of archaeological fieldwork and the allocation of documentation resources in the North Sea region. Sites that preserve organic material are at greater and more immediate risk than those preserving only stone, brick, or iron, and the risk is accelerating. In practical terms, this means that a site documented adequately by remote sensing in 2010 may by 2035 have lost the organic material that makes it archaeologically significant.

We propose a tiered response framework with three levels of urgency:

| Tier | Risk level                                           | Recommended action                             |
|------|------------------------------------------------------|------------------------------------------------|
| 1    | Exposed organic material, active erosion             | Urgent systematic documentation within 3 years |
| 2    | Buried organic material, evidence of recent exposure | Systematic documentation within 5-7 years      |
| 3    | Buried organic material, stable sediment             | Monitoring and risk assessment every 10 years  |

### 5. Research Programme and EU Application

The EU Horizon grant application 101187724, submitted in February 2026, proposes a systematic programme of accelerated documentation and sediment monitoring at high-priority heritage sites in the German Bight and Scottish waters, using the North Frisian Survey methodology as a model. The programme pairs archaeological documentation with oceanographic monitoring to create the first long-term dataset linking sediment dynamics to organic preservation rates at wreck sites. Caledonia Maritime University will lead the Scottish component and the Nordmeer Maritime University team will lead the German Bight component.

This working paper will be updated with peer review comments and submitted to a journal in the second half of 2026. The data from NF-7 discussed in this paper are subject to the project publication embargo and should not be cited without the PI's permission.